# Data Science

## Lecture 10: Projections and Dimensionality Reduction

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# Where We Left Off

Lectures 08–09 found structure in more dimensions than anyone can look at. Today maps them into two — which requires deciding what to lose.

**Lecture 07**
plot it

→

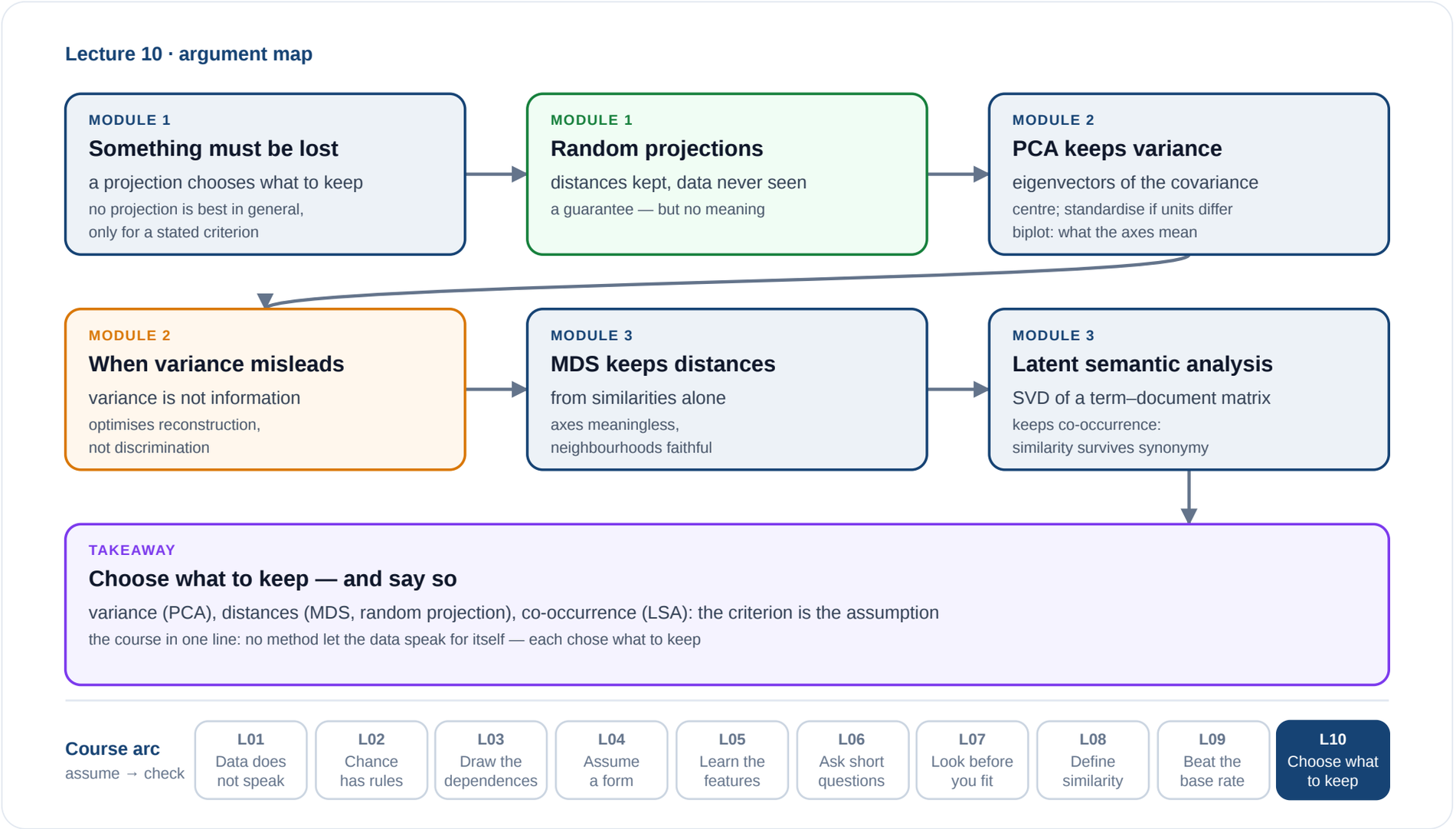
**Lectures 08–09**
too many dimensions to plot

→

**Today**
project: choose what to keep

> This is the last lecture — and the last instance of Lecture 01's claim: no projection lets the data speak for itself.

# Argument Map



# Module 1

## What a Projection Preserves

> ***Guiding Question:*** *Reducing dimensions must lose information. Which information do you choose to keep?*

# Seeing High-Dimensional Data

A **geometric projection** maps high-dimensional data onto a low-dimensional space — usually 2-D or 3-D, so that it can be plotted and reasoned about.

## Linear projections

$$\mathbf{z} = \mathbf{P}\mathbf{x}$$

A matrix $\mathbf{P}$ maps every point. Distortion is global and predictable, and each new axis is a weighted sum of the old ones. A *reducing* $\mathbf{P}$ is not invertible: what it drops cannot be recovered.

## Non-linear projections

No single matrix. Methods such as **t-SNE** and **UMAP** keep **local neighbourhoods** — who is near whom — at the price of global distances. Often more faithful visually, harder to interpret.

**The central question of the chapter:** how do we find the *best* projection — and by which criterion is one projection better than another?

# What Is Necessarily Lost

Reducing dimensions **must** lose information — the question is only which.

### Variance

PCA keeps as much of it as a linear map can.

### Distances

MDS keeps pairwise distances as faithfully as possible.

### Guarantees

Random projection keeps distances within a provable error, cheaply.

A projection is defined by **what it chooses to preserve**. There is no best-in-general projection, only a best one for a stated criterion.

# The Surprising Result

**Johnson–Lindenstrauss lemma.** For any $0 < \epsilon < 1$ and $n$ points in any dimension, a projection onto

$$d = O\left(\frac{\log n}{\epsilon^2}\right)$$

dimensions exists that preserves all pairwise distances to within a factor $1 \pm \epsilon$.

The target dimension depends on the **number of points** and on the accuracy wanted — **not on the original dimensionality**. A million-dimensional dataset and a thousand-dimensional one need the same target dimension.

# Construction

1. Draw a $d \times p$ matrix $\mathbf{R}$ with entries sampled independently (Gaussian, or $\pm 1$ with equal probability).
2. Project: $\mathbf{z} = \frac{1}{\sqrt{d}}\mathbf{R}\mathbf{x}$.

With $d$ as in the lemma, such a random $\mathbf{R}$ meets the $1 \pm \epsilon$ bound **with high probability** — the lemma's projection can simply be drawn.

# Properties and Limits

## Properties

No training, no data pass — the matrix does not depend on the data at all. Extremely fast, trivially applied to new points, and it comes with a bound.

## Limits

The axes are meaningless, so the result is not interpretable. It preserves distances, not structure, and works best when $d$ is not too small.

Random projection preserves distances **with a guarantee and without looking at the data**. Use it when you need speed and a bound; use PCA when you need meaning.

# Module 2

## Principal Component Analysis

> ***Guiding Question:*** *PCA keeps the directions of largest variance. When is variance the wrong thing to keep?*

# Which Direction Matters?

A cloud stretched along a diagonal: projected onto the **short** axis it collapses into a blob; onto the **long** axis its points stay apart.

**PCA's criterion:** the 1st component is the direction of **maximum variance**; each further one takes the most *remaining* variance, **orthogonal** to all before it.

The first $m$ components are the **best $m$-dimensional linear approximation**: most variance kept = least squared reconstruction error.

PCA keeps the directions of highest **variance**. Its assumption — variance is information — makes it powerful, and makes it fail.

# Covariance Is the Object to Diagonalise

For centred data $\mathbf{X}$, the **covariance matrix**

$$\mathbf{C} = \frac{1}{n-1}\mathbf{X}^\top\mathbf{X}$$

holds the variance of each attribute on its diagonal and the covariance of each pair off it.

### Large diagonal entry

That attribute varies a lot — potentially informative.

### Large off-diagonal entry

Two attributes are **redundant**: one partly predicts the other.

The goal: find new axes in which the covariance matrix is **diagonal** — no redundancy left between components — and order them by variance.

# The Solution

The directions that diagonalise $\mathbf{C}$ are its **eigenvectors**; the variance captured by each is the corresponding **eigenvalue**.

Sorting eigenvectors by descending eigenvalue gives the principal components, and the projection matrix $\mathbf{W}_m$ holds the top $m$ of them.

$\mathbf{C}$ is symmetric, so its eigenvectors are orthogonal: the components are **uncorrelated by construction**. The share of total variance in component $i$ is its **explained variance** $\lambda_i / \sum_j \lambda_j$.

# A Two-Attribute Example

Two attributes in the same unit, each with variance 2, covariance 1:

$$\mathbf{C} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad \mathbf{C}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = 3\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad \mathbf{C}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = 1\begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

| component | direction | eigenvalue | explained variance |
|---|---|---|---|
| PC1 | $(1,1)/\sqrt{2}$ — "both together" | $\lambda_1 = 3$ | $3/4 = 75\%$ |
| PC2 | $(1,-1)/\sqrt{2}$ — "one against the other" | $\lambda_2 = 1$ | $1/4 = 25\%$ |

PCA is the **eigendecomposition of the covariance matrix**: eigenvectors give the directions, eigenvalues give their importance, and the top ones are kept.

# The Procedure

1. **Centre** the data; **standardise** it if the units differ.
2. Compute the covariance matrix $\mathbf{C}$.
3. Compute its eigenvectors and eigenvalues.
4. Sort by descending eigenvalue; keep the top $m$.
5. Project: $\mathbf{Z} = \mathbf{X}\mathbf{W}_m$.

**Centring is not optional**: the covariance formula assumes it. **Standardising is a choice** — make it when units differ. Otherwise an attribute in metres dominates one in kilometres purely through its **unit**, and PCA reports the choice of unit as the principal direction.

# Choosing the Number of Components

A **scree plot** shows eigenvalues in descending order; the usual reading is the same elbow heuristic as for choosing $k$ in clustering, or a cumulative explained-variance threshold (80%, 95%).

### Benefits

Removes correlated and noisy directions; speeds up downstream methods; makes 2-D visualisation possible; components are uncorrelated by construction.

### Drawbacks

Components are **linear combinations of all attributes**, so interpretability is lost. Only **linear** structure is captured. And variance is assumed to equal importance — which for classification can be false.

# Mini-Exercise

You run PCA on the standardised iris data; the first two components explain 96% of the variance.

**Are these two components therefore good features for telling the species apart?**

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> PCA is centre → covariance → eigendecomposition → keep the top components. It optimises **reconstruction**, not discrimination; centring is mandatory, standardising a choice made by the units.

# Points and Variables in One Picture

A **biplot** overlays two things on the same axes:

### The objects

Each data point, plotted by its scores on the first two components — an ordinary scatterplot of the projection.

### The variables

Each original attribute drawn as an **arrow**: its loading on the two components. Direction shows what the attribute contributes to, length shows how strongly.

A biplot is how you recover interpretability after PCA has destroyed it: the arrows say what the anonymous components *mean* in terms of the original attributes.

# Reading the Arrows

| arrows | the two attributes are |
|---|---|
| pointing together | positively correlated |
| pointing in opposite directions | negatively correlated |
| at about 90° | roughly uncorrelated |
| long | well represented by the two components |
| short | mostly outside the plotted plane — do not read its angles |

The biplot answers "what do these axes mean?" by drawing the original attributes into the projected space — points and variables read together.

# The Churn Shop: What Does PC1 Mean?

Four customer attributes: tenure, annual spend, app logins, support tickets. A biplot of the standardised table might show:

### THE ARROWS

Tenure, spend and logins point **together** along PC1; tickets points the **other way**.

### THE READING

PC1 is something like **engagement**, and the customers who churned sit at its low end.

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**THE READING**

PC1 is something like **engagement**, and the customers who churned sit at its low end.

Unstandardised, spend in euros has by far the largest variance, and PC1 is simply **spend**. Standardising decides what PC1 can mean — Lecture 08's scaling choice again.

# Module 3

## Beyond Variance

> ***Guiding Question:*** *Some objects have no coordinates — only similarities. Can they still be mapped?*

# A Different Criterion

PCA needs coordinates and preserves **variance**. MDS needs only a **dissimilarity matrix** and preserves **the distances themselves**.

Given pairwise dissimilarities $d_{ij}$, find low-dimensional points $\mathbf{z}_1, \ldots, \mathbf{z}_n$ minimising Kruskal's **stress-1**

$$S = \sqrt{\frac{\sum_{i<j} \left( \|\mathbf{z}_i - \mathbf{z}_j\| - d_{ij} \right)^2}{\sum_{i<j} d_{ij}^2}}$$

This is the projection method for objects that have **no coordinates at all** — only a dissimilarity: documents, molecules, survey respondents, perceived similarity ratings turned into distances.

# Classical MDS and Its Relatives

### Classical (metric) MDS

Solved in closed form by an eigendecomposition of the (double-centred) squared distance matrix. With Euclidean distances it is **equivalent to PCA**.

### Non-metric MDS

Preserves only the **rank order** of distances. Suited to ordinal similarity data — "A is more like B than like C" — where the numbers are not meaningful.

# Reading the Result

### Stress tells you the cost

A 2-D map of genuinely high-dimensional structure has high stress: the picture is a compromise, and the number says how big a one.

### Axes mean nothing

Unlike PCA components, MDS axes have no interpretation — only **relative positions** carry information. Rotations and reflections are equivalent.

MDS preserves **distances** rather than variance, works from a dissimilarity matrix alone, and produces a map whose axes are meaningless but whose **neighbourhoods are faithful** — as far as the stress allows.

# The Problem With Matching Words

A document is represented as a vector of term counts — the **term–document matrix**. Similarity is then the overlap of vocabulary, which fails twice:

### Synonymy

"car" and "automobile" never match, though the documents are about the same thing.

### Polysemy

"bank" matches across finance and rivers, though the documents are unrelated.

> The vocabulary is the wrong space. We want a space of **concepts**, in which related terms lie close together.

# The Approach

Apply a **singular value decomposition** to the term–document matrix $\mathbf{A}$:

$$\mathbf{A} \approx \mathbf{U}_k \mathbf{\Sigma}_k \mathbf{V}_k^\top$$

and keep the $k$ largest singular values. Documents and terms are both represented in the resulting $k$-dimensional **latent space**.

SVD is PCA's underlying machinery applied to a rectangular matrix — the same "keep the strongest directions" idea, on term–document data (usually without centring).

# What the Latent Space Buys

### Dimension reduction

Thousands of terms collapse to a few hundred latent dimensions.

### Synonyms merge

Terms that co-occur with the same contexts end up close together.

### Noise drops

Rare, idiosyncratic usage lives in the discarded small singular values.

LSA projects documents and terms into a shared **latent space** via SVD, so that similarity survives synonymy. It is dimensionality reduction used as a representation, not just as compression.